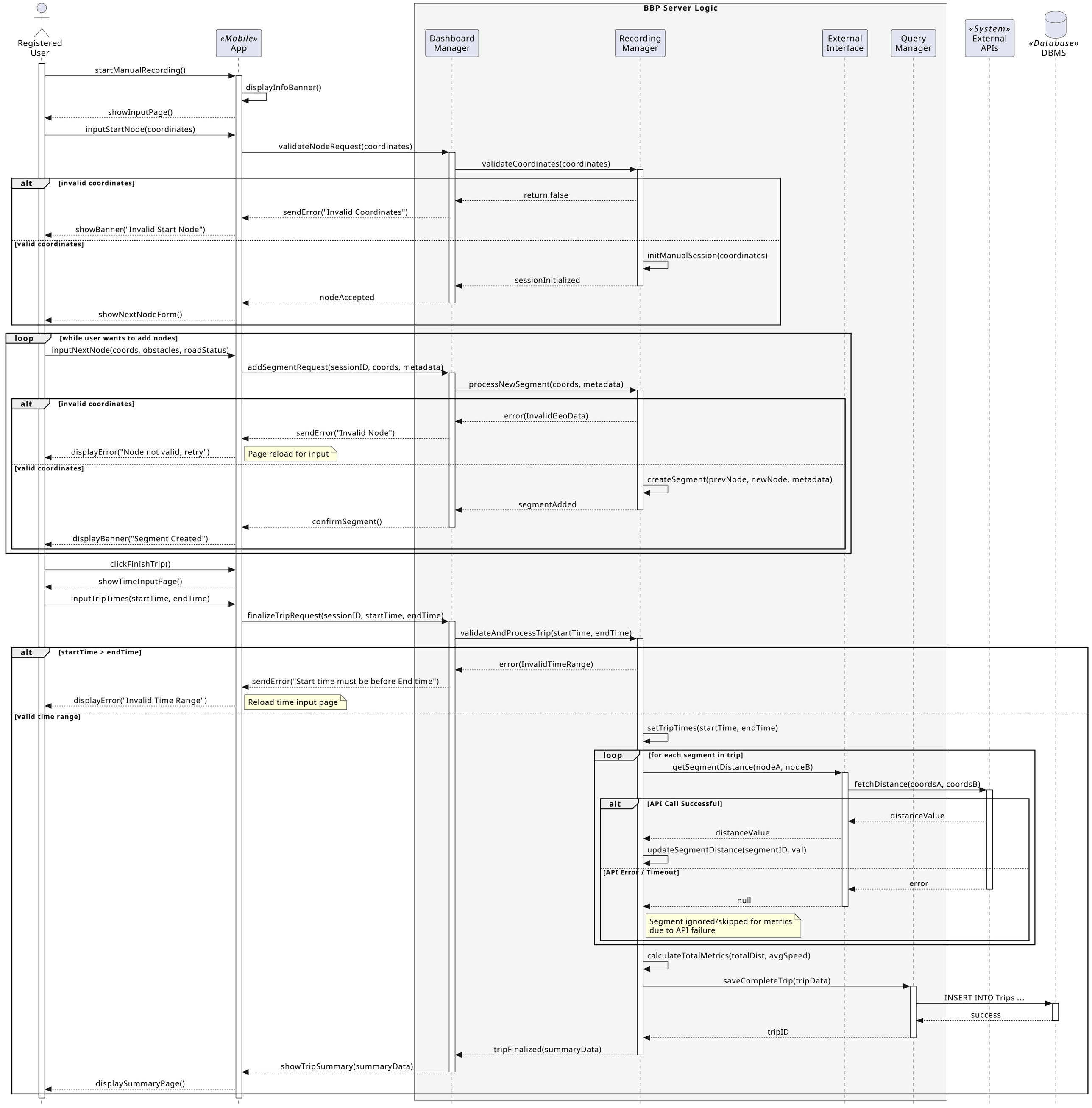
BBP Server Logic
Registered User
«Mobile» App
Dashboard Manager
Recording Manager
External Interface
Query Manager
«System» External APIs
«Database» DBMS
startManualRecording()
displayInfoBanner()
showInputPage()
inputStartNode(coordinates)
validateNodeRequest(coordinates)
validateCoordinates(coordinates)
alt [invalid coordinates]
return false
sendError("Invalid Coordinates")
showBanner("Invalid Start Node")
[valid coordinates]
initManualSession(coordinates)
sessionInitialized
nodeAccepted
showNextNodeForm()
loop [while user wants to add nodes]
inputNextNode(coords, obstacles, roadStatus)
addSegmentRequest(sessionID, coords, metadata)
processNewSegment(coords, metadata)
alt [invalid coordinates]
error(InvalidGeoData)
sendError("Invalid Node")
displayError("Node not valid, retry")
Page reload for input
[valid coordinates]
createSegment(prevNode, newNode, metadata)
segmentAdded
confirmSegment()
displayBanner("Segment Created")
clickFinishTrip()
showTimeInputPage()
inputTripTimes(startTime, endTime)
finalizeTripRequest(sessionID, startTime, endTime)
validateAndProcessTrip(startTime, endTime)
alt [startTime > endTime]
error(InvalidTimeRange)
sendError("Start time must be before End time")
displayError("Invalid Time Range")
Reload time input page
[valid time range]
setTripTimes(startTime, endTime)
loop [for each segment in trip]
getSegmentDistance(nodeA, nodeB)
fetchDistance(coordsA, coordsB)
alt [API Call Successful]
distanceValue
distanceValue
updateSegmentDistance(segmentID, val)
[API Error / Timeout]
error
null
Segment ignored/skipped for metrics due to API failure
calculateTotalMetrics(totalDist, avgSpeed)
saveCompleteTrip(tripData)
INSERT INTO Trips ...
success
tripID
tripFinalized(summaryData)
showTripSummary(summaryData)
displaySummaryPage()